

assignment kine1D_2

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:52pm

Quiz Instructions

2 attempts - see tutorial in dropbox. After you try hard



Question 1 10 pts

A car is traveling with a constant speed when the driver suddenly applies the brakes, causing the car to slow down with a constant acceleration of magnitude 3.50 m/s^2 . If the car comes to a stop in a distance of 30.0 m, what was the car's original speed?

☐

210 m/s

☐

105 m/s

☐

315 m/s

☐

10.2 m/s

☐

14.5 m/s



Question 2 10 pts

A runner maintains constant acceleration after starting from rest as she runs a distance of 60.0 m. The runner's speed at the end of the 60.0 m is 9.00 m/s. How much time did it take the runner to complete the 60.0 m distance?

☐

15.0 s

☐

9.80 s

☐

10.2 s

☐

13.3 s

☐

6.67 s



Question 3 10 pts

A car starts from rest and accelerates with a constant acceleration of 1.00 m/s^2 for 3.00 s . The car continues for 5.00 s at constant velocity. How far has the car traveled from its starting point?

☐

19.5 m

☐

24.0 m

☐

9.00 m

☐

4.50 m

☐

15.0 m



Question 4 10 pts

A ball rolls across a floor with an acceleration of 0.100 m/s^2 in a direction opposite to its velocity. The ball has a velocity of 4.00 m/s after rolling a distance 6.00 m across the floor. What was the initial speed of the ball?

☐

3.85 m/s

☐

4.60 m/s

☐

5.85 m/s

☐

5.21 m/s

☐

4.15 m/s



Question 5 10 pts

A car is traveling at 26.0 m/s when the driver suddenly applies the brakes, causing the car to slow down with constant acceleration. The car comes to a stop in a distance of 120 m . How fast was the car moving when it was 60.0 m past the point where the brakes were applied?

☐

18.4 m/s

☐

12.1 m/s

☐

9.20 m/s



22.5 m/s



15.0 m/s



Question 6 10 pts

An object starts from rest at time $t = 0.00$ s and moves in the $+x$ direction with constant acceleration. The object travels 12.0 m from time $t = 1.00$ s to time $t = 2.00$ s. What is the acceleration of the object?

The tutorial is not clear here. Try first without looking at hint.

hint: you don't know the speed at $t=1$ s. It is not 0.

at $t = 1$ s, you have $x_1 = 0.5 (a) (t^2) = 0.5 (a) (1)$

at $t = 2$ s, you have $x_2 = 0.5 (a) (t^2) = 0.5 (a) (4)$

$x_2 - x_1 = 12 = 2a - a/2$.

8.00 m/s²4.00 m/s²-12.0 m/s²24.0 m/s²-4.00 m/s²

Question 7 10 pts

A car accelerates from 10m/s to 30m/s at a rate of 3m/s/s How far does the car travel while accelerating?



80.0 m



399 m



226 m



133 m



Question 8 10 pts

The average speed of a horse that gallops 10 kilometers in 30 minutes is



20 km/h.



40 km/h.



15 km/h.



30 km/h.



Question 9 10 pts

A dragster starts from rest and travels 1/4 mi in 6.70 s with constant acceleration. What is its velocity when it crosses the finish line?



135 mi/h



296 mi/h



269 mi/h



188 mi/h



Question 10 10 pts

A racquetball strikes a wall with a speed of 30 m/s and rebounds in the opposite direction with a speed of 26 m/s. The collision takes 20 ms. What is the magnitude of average acceleration of the ball during the collision with the wall?

(the magnitude of the acceleration is asked. so do not include the negative sign)

1500 m/s²200 m/s²2800 m/s²1300 m/s²



0 m/s²

Not saved

Submit Quiz